

Cyclistic Bike-Share

How riders use the bikes, and where Cyclistic should grow · 663K trips · NYC · 2019–2020

663K

Trips analysed

+26.6%

YoY growth

$r = 0.66$

Temp ↔ demand

91%

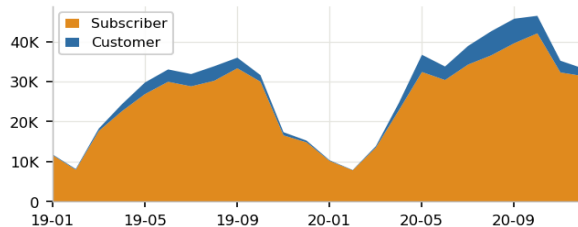
Member trips

The situation

Cyclistic's product team needs to understand how riders use the bikes before funding new stations. The core question: how do annual **members** differ from casual **customers**, and what drives demand across season, weather, and location? I queried and cleaned 663,084 trips in BigQuery, then built an interactive dashboard and this analysis to answer it.

What the data shows

Ridership grew +26.6%, with a COVID dip

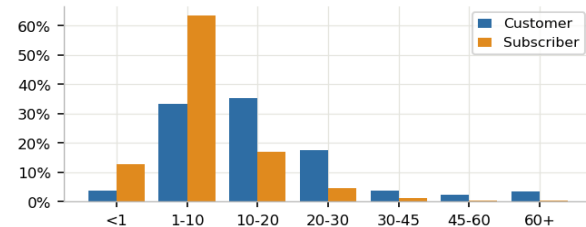


Strong seasonality; members (orange) carry volume, casuals (blue) widen in summer.

Plus: weather is the top demand driver (daily trips correlate **+0.66** with temperature, rain cuts demand ~20%), and demand concentrates in a few Manhattan ZIP codes — the clearest sites for new capacity.

The dashboard: an interactive three-tab workbook. Stakeholders can filter every chart at once by rider type, date range, and area; switch tabs between time, location, and usage views; and hover any point for exact trip counts, durations, and weather values.

Members and casuals ride differently



64% of member trips are under 10 min (commute); casuals spread longer (leisure).

Recommendations

- **Convert casuals with a weekend- and summer-framed membership offer** — targeting them when leisure ridership peaks.
- **Forecast and rebalance the fleet using daily weather**, not just the calendar, given the strong temperature and rain effects.
- **Add stations in the high-demand Manhattan core first**, then expand toward growing Brooklyn areas; run maintenance in shoulder seasons to protect the summer peak.

Skills demonstrated

BigQuery SQL · weighted aggregation & data-integrity checks (sentinel detection, outlier handling) · correlation analysis · dashboard design · stacked-area, scatter, distribution & ranked-bar visualisation · stakeholder-driven analysis · translating findings into business recommendations.

Data: NYC Citi Bike Trips, NOAA GSOD weather & US Census boundaries (public datasets), joined in BigQuery. Full methodology and supporting charts follow on pages 2–4.

01 Data and method

The dataset is **59,730 aggregated rows representing 663,084 individual trips**. Each row is a unique combination of rider type, start and end location (ZIP, neighbourhood, borough), date, daily weather, and trip duration, with a **trip_count** weight. Because rows are pre-grouped, every statistic in this report is **weighted by trip_count** — a plain average of the rows would misrepresent the trips.

Data sources

The analysis table was assembled in BigQuery by joining four public / shared sources:

Source	Contributes
NYC Citi Bike Trips (BigQuery public dataset)	Trip records: rider type, start/stop, ZIP, duration, count
Census Bureau US Boundaries (TIGER/Line)	Borough & neighbourhood geography, joined to trip ZIPs
GSOD — NOAA Global Surface Summary of the Day	Daily temperature, wind speed, precipitation, joined on date
ZIP-code spreadsheet (shared)	ZIP → neighbourhood lookup

Note: the 999.9 wind values flagged below are GSOD's documented missing-data code, not real readings.

Data-quality checks performed

- **Duration outliers:** 124 trips (0.02%) exceeded 24 hours and were removed as implausible. The 79,049 trips under one minute (11.9%) were kept — they are genuine short hops, and the duration field is quantised to ~10-minute steps, so very short trips legitimately round toward zero.
- **Weather sentinel:** day_mean_wind_speed contained a 999.9 placeholder on 11 days (4,705 trips); these were treated as missing in weather analysis. Real wind ranges 0.7–13 mph.
- **Integrity:** no nulls, no duplicate rows. 2.9% of trips cross boroughs; 42.5% start and end in the same neighbourhood.

02 Headline findings

Finding	Evidence
Ridership grew +26.6% year-over-year	292,505 trips in 2019 → 370,455 in 2020
Demand tracks temperature strongly	Correlation $r = +0.66$ (daily trips vs. mean temp)
Rain cuts daily ridership ~20%	976 trips/day dry vs. 785 wet (>0.1 in)
Members dominate volume	91% of trips are Subscribers (606K vs. 57K)
Two distinct usage patterns	Members 12.6 min midweek; casuals 24.8 min weekends
COVID-19 dip then rebound	–24% in Mar 2020 vs. 2019; +23% by May 2020

03 Low-fidelity mockup

Before building the charts, I sketched a low-fidelity wireframe to plan the layout and components — a title bar and tab row, a filter strip, a row of KPI cards, two chart zones, and a map plus a top-traffic table at the bottom. The dashed placeholders fix **what** goes where before any real data is drawn, which keeps the build focused and the final layout consistent.

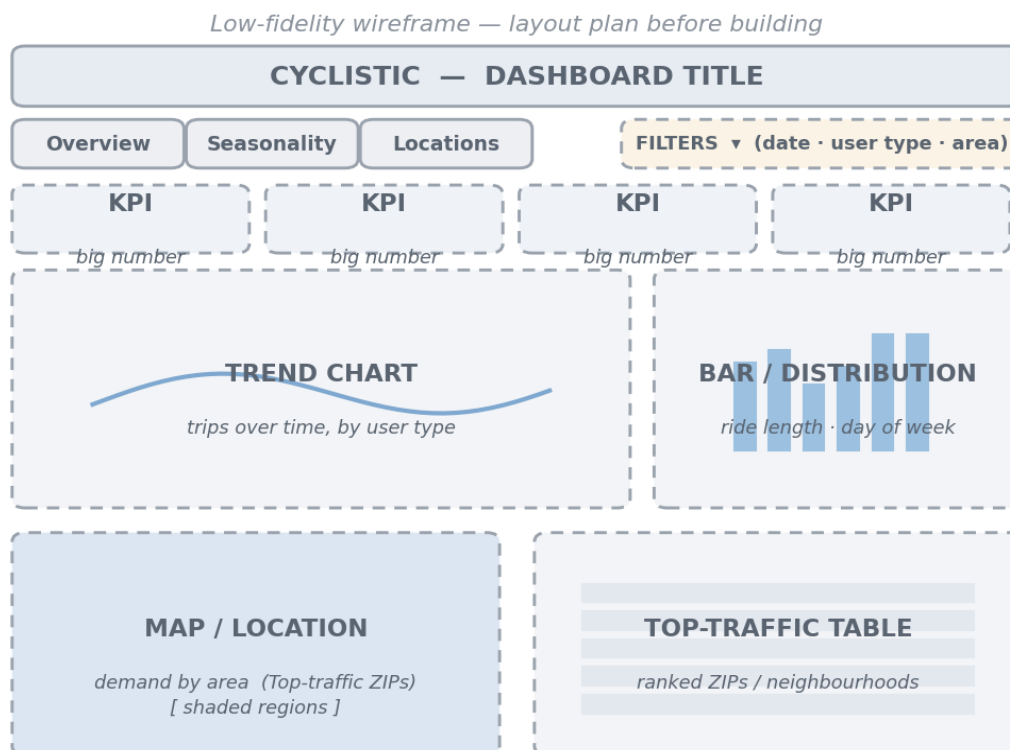


Figure 1 — Low-fidelity wireframe: planned dashboard components and layout.

04 The built dashboard

The wireframe above was built out into the interactive workbook below. The final layout follows the plan — title bar and tabs, a KPI row, the trend and distribution charts, the demand-vs-temperature view, and the top-traffic ranking — with live filters that apply across every chart.

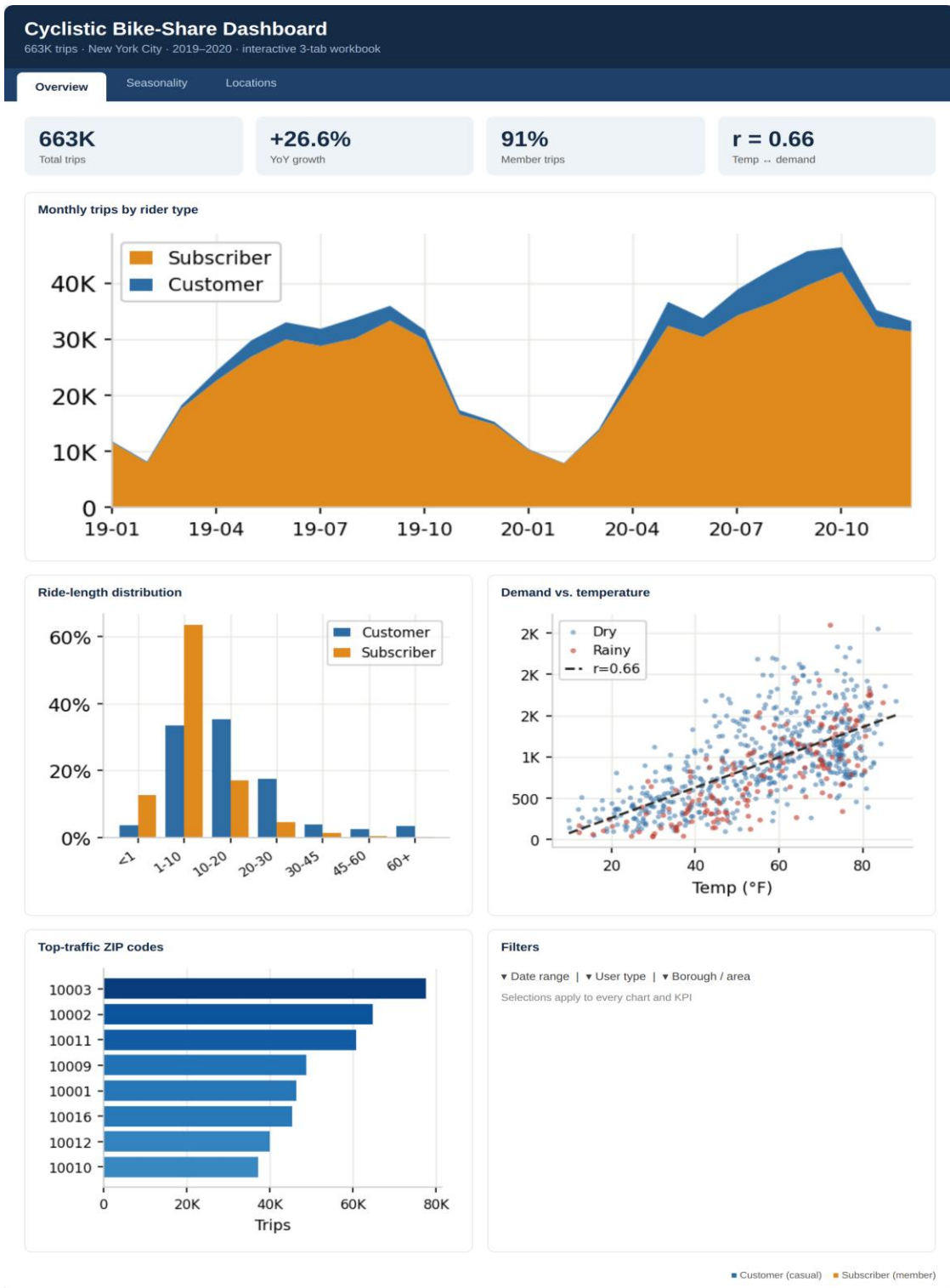


Figure 2 — Snapshot of the built dashboard (Overview tab). Charts update live as filters are applied.

05 Finding 1 · Growth and seasonality

Monthly ridership follows a strong seasonal wave — peaking each summer, bottoming in winter — and grew **+26.6% from 2019 to 2020**. The spring-2020 COVID-19 lockdown is visible as a sharp dip in March (−24% versus 2019) followed by a rapid rebound, with 2020's second half exceeding all of 2019. Subscribers (orange) carry the bulk of volume in every month; the casual band (blue) widens in summer.

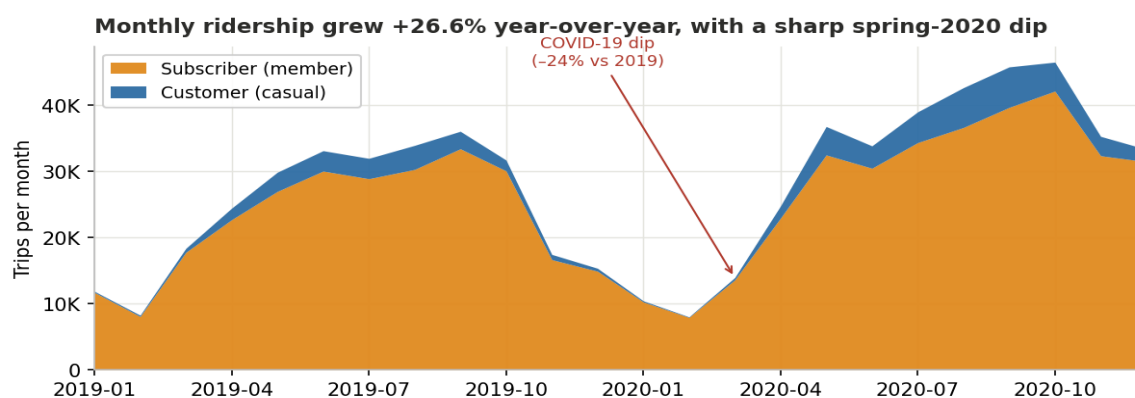


Figure 3 — Monthly trips by rider type, with year-over-year growth and the COVID-19 dip annotated.

06 Finding 2 · Weather is a primary demand driver

Plotting each day's total trips against its mean temperature gives a clear positive relationship ($r = 0.66$). Cold days ($<32^{\circ}\text{F}$) average ~330 trips; warm days ($75\text{--}90^{\circ}\text{F}$) average ~1,270 — nearly 4×. Rainy days (red) sit visibly below the trend, confirming a separate ~20% rain penalty. Note this is correlational — warm months also bring longer daylight and more tourism — but weather is the strongest single external signal in the data.

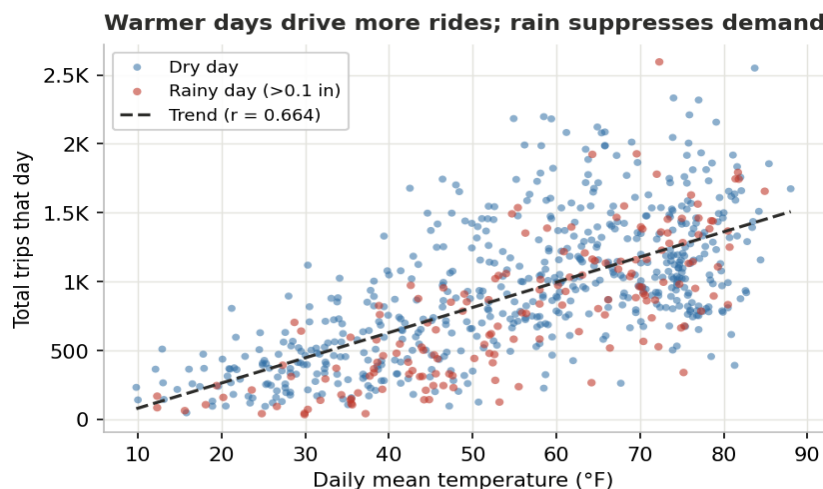


Figure 4 — Daily ridership vs. temperature, split by dry/rainy days, with fitted trend and correlation.

07 Finding 3 · Members and casuals ride differently

Normalising trip length to each group's share of trips exposes the behavioural split clearly. **64% of member trips last 1–10 minutes** — short, uniform, commute-shaped. Casual trips spread much wider, with over half running 10 minutes or longer and a meaningful tail past 30 minutes — the signature of leisure and exploratory riding.

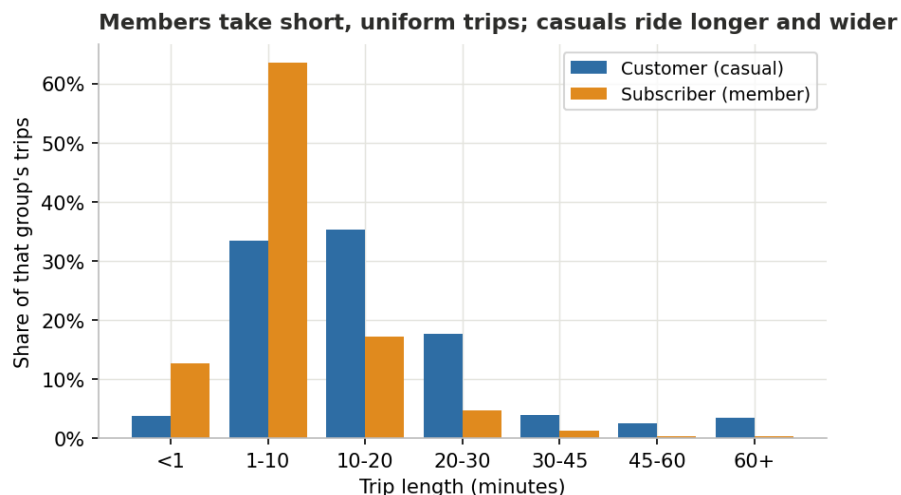


Figure 5 — Trip-length distribution as a share of each group's own trips.

The weekly pattern reinforces this. Members peak Monday–Wednesday and dip on weekends (only **25% of their trips fall on weekends**, below the 28.6% baseline). Casuals do the opposite, with **39% of trips on weekends** — a clear leisure profile.

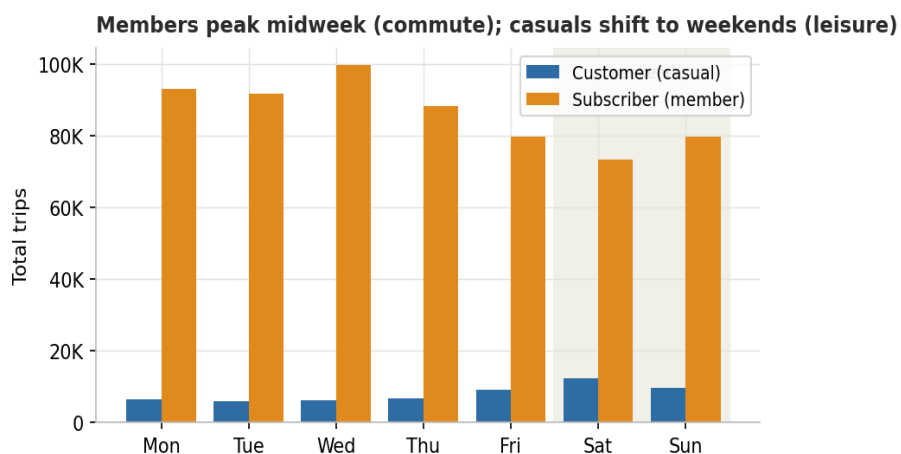


Figure 6 — Trips by day of week; weekend shaded. Members commute, casuals ride weekends.

08 Finding 4 · Where the traffic concentrates

Ranking pickup locations shows demand is heavily concentrated in Lower and Midtown Manhattan. The busiest ZIP codes — 10003, 10002 and 10009 (Lower East Side), and 10011 / 10001 (Chelsea & Clinton) — each generate 45,000–78,000 trips, and the top twelve ZIPs account for **78% of all trips**. These are the clearest candidates for added station capacity.

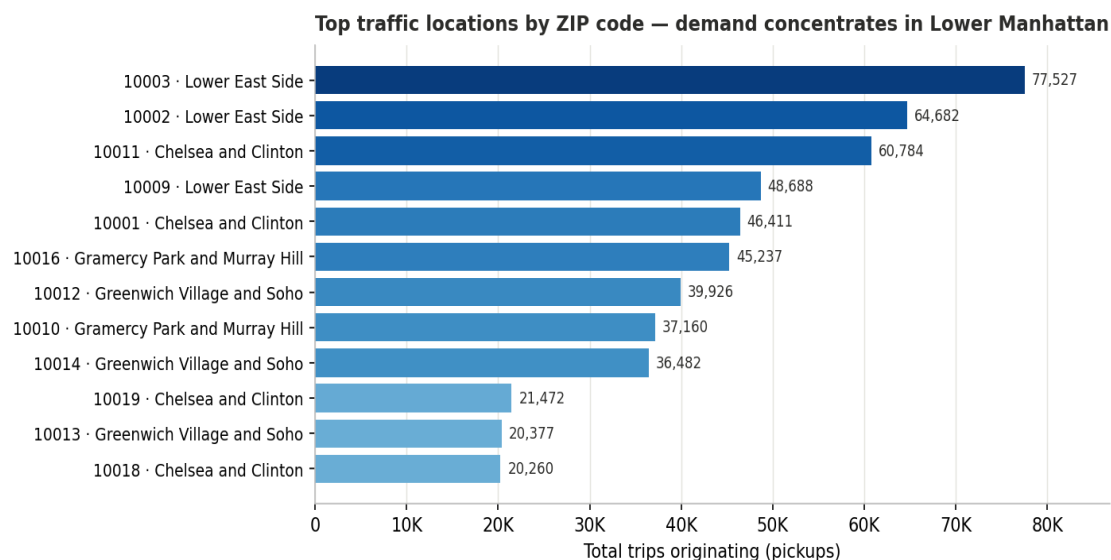


Figure 7 — Top traffic locations by ZIP code (with neighbourhood), ranked by trips originating.

Top 10 highest-traffic locations (by trips originating):

#	ZIP	Neighbourhood	Borough	Trips	Share
1	10003	Lower East Side	Manhattan	77,527	11.7%
2	10002	Lower East Side	Manhattan	64,682	9.8%
3	10011	Chelsea and Clinton	Manhattan	60,784	9.2%
4	10009	Lower East Side	Manhattan	48,688	7.3%
5	10001	Chelsea and Clinton	Manhattan	46,411	7.0%
6	10016	Gramercy Park & Murray Hill	Manhattan	45,237	6.8%
7	10012	Greenwich Village & Soho	Manhattan	39,926	6.0%
8	10010	Gramercy Park & Murray Hill	Manhattan	37,160	5.6%
9	10014	Greenwich Village & Soho	Manhattan	36,482	5.5%
10	10019	Chelsea and Clinton	Manhattan	21,472	3.2%

Data note: the source is aggregated to ZIP-code level, not individual docking stations, so traffic is reported by ZIP / neighbourhood rather than by named station.

09 Dashboard capabilities

The accompanying workbook is an interactive, three-tab dashboard — not a static report. Stakeholders can:

- **Filter the entire view by rider type** (All / Subscriber / Customer); every chart and KPI redraws together, so members and casuals can be compared side by side in one click.

- **Switch tabs** between Seasonality (trend + heatmap), Summer trends (location demand), and Top trips (trip-minute leaders) to move between time, place, and usage views.
- **Hover any data point** to read exact values — trip counts, average duration, and the underlying weather metric — rather than estimating from the axis.
- **Read the neighbourhood × month heatmap** to spot exactly when and where demand peaks, and use the ranked bars to identify the highest-traffic ZIPs at a glance.

10 What this means for Cyclistic

- **Convert casuals where the behaviour gap is widest.** Casual riders look like weekend leisure users; a weekend- and summer-framed membership offer targets them when they are most active.
- **Plan capacity around weather, not just the calendar.** With $r = 0.66$ on temperature and a 20% rain penalty, demand forecasting and rebalancing should weight daily weather, concentrating fleet on warm dry days.
- **Protect the summer peak.** Maintenance and redistribution should run in shoulder seasons so the fleet is fully available through the summer demand crest.